

Haqian Han , M.Sc. Student

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About:

I am a M.Sc. student at Tsinghua University, supervised by **Prof. Xiangyang Ji**. My research focuses on bio-inspired cameras and fundamental computer vision science. I have published some papers, all with open-source code.

Research Interests:

Computational Imaging; Bio-inspired Cameras; Simulation

Education

2026 – 2030 (expected)	Ph.D.	 Northwestern University	Evanston, IL, USA
		Computer Science - Advisor: Prof. Emma Alexander	
2023 – 2026	M.Sc.	 Tsinghua University	Beijing, China
		Control Science and Engineering - Advisor: Prof. Xiangyang Ji - GPA: 3.64/4.00	
		Relevant Coursework: Computer Graphics, Matrix Analysis and Applications, Convex Optimization, Information Theory	
2019 – 2023	B.Eng.	 Zhejiang University	Hangzhou, China
		Robotics Engineering - Chu Kochen Honors College - GPA: 3.84/4.00	
		Relevant Coursework: Mathematical Analysis II (H), Linear Algebra II (H), General Physics II (H), Probability Theory and Statistics	

Publications

1. H. Han, J. Lyu, J. Li, et al. **European Conference on Computer Vision 2024**
Physical-Based Event Camera Simulator 
This paper presents a physics-based method that simulates the complete optical path of event cameras, intended for fundamental academic research in event camera modeling and evaluation.
2. H. Han, J. Li, H. Wei, et al. **Annual Conference on Neural Information Processing Systems 2024**
Event-3DGS: Event-based 3D Reconstruction Using 3D Gaussian Splatting 
This research proposes a computational method for three-dimensional visual scene modeling from bio-inspired camera, with civilian applications in consumer virtual reality and smartphone imaging.
3. J. Li, Y. Zhang, H. Han, and X. Ji **IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025**
Active Event-based Stereo Vision 
This paper presents a depth-measurement system using a pair of synchronized bio-inspired camera, enabling better three-dimensional distance estimation.
4. J. Li, Y. Zhang, H. Han, K. Huang, and X. Ji **IEEE Transactions on Pattern Analysis and Machine Intelligence 2026**
Towards Ultrafast Depth Sensing Via Active Event-Based Stereo Vision 
This extension optimizes the computational architecture for depth sensing, designed specifically to reduce motion blur and improve focus speed.

5. H. Han, et al.

arXiv preprint

Learning to Remove Lens Flare in Event Camera (preprint)



This paper develops a physics-based computational framework to detect and eliminate flare caused by bright light sources in event camera, improving image quality for civilian photography applications.

6. Z. Song, H. Lu, H. Han, B. Li, et al.

European Conference on Computer Vision 2026

Degradation-Agnostic Clarity Learning for Unpaired Image Dehazing



This paper introduces a degradation-agnostic learning framework for unpaired image dehazing, improving restoration quality across diverse real-world haze conditions.

Internship Experience



CNRS Singapore Singapore • Research Intern

Summer 2025

Conducted research on lens flare in event camera. Led the first systematic study of lens flare in event cameras and developed a removal framework.

Teaching Assistant

2025 & 2026

■ **Foundations of Machine Learning**, Tsinghua University.

Instructor: Prof. Xiangyang Ji

Duties: Graded written homework assignments and tabulated scores for graduate students.

2024

■ **Computer Vision**, Tsinghua University.

Instructor: Prof. Xiangyang Ji

Duties: Graded written homework assignments and tabulated scores for undergraduate students.

Awards & Achievements

2026

■ **Outstanding Master's Thesis Award**, Tsinghua University.

2025

■ **Comprehensive Excellence Scholarship**, Tsinghua University.

2024

■ **Comprehensive Excellence Scholarship**, Tsinghua University.

■ **Hybrid Teaching Environment and Application Test Pass Certificate**, Tsinghua University.

2023

■ **Outstanding Graduate**, Zhejiang University.

2022

■ **Academic Excellence Award**, Zhejiang University.

■ **Student Innovation and Entrepreneurship Award**, Zhejiang University.

2021

■ **Academic Excellence Award**, Zhejiang University.

2020

■ **Academic Excellence Award**, Zhejiang University.

Skills

Programming

■ Python, C++, MATLAB.

Languages

■ Mandarin Chinese (Native), English (Proficient).